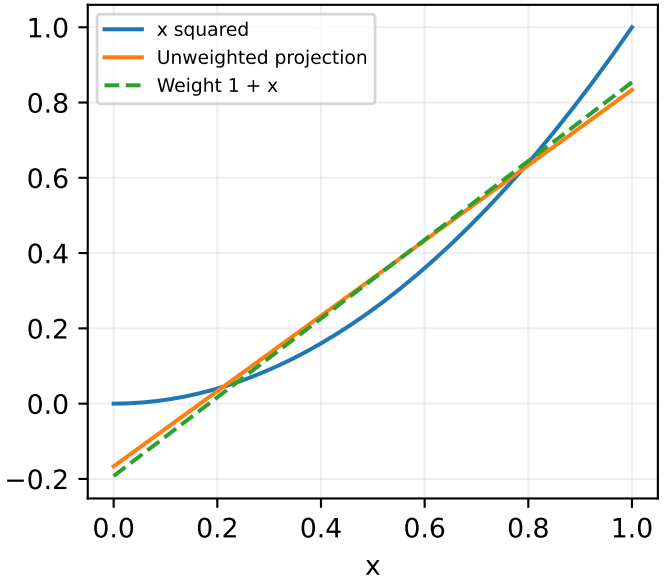
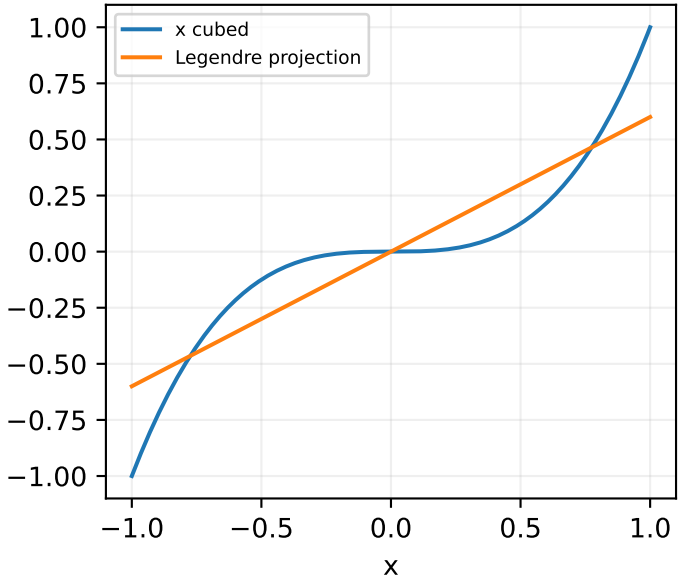
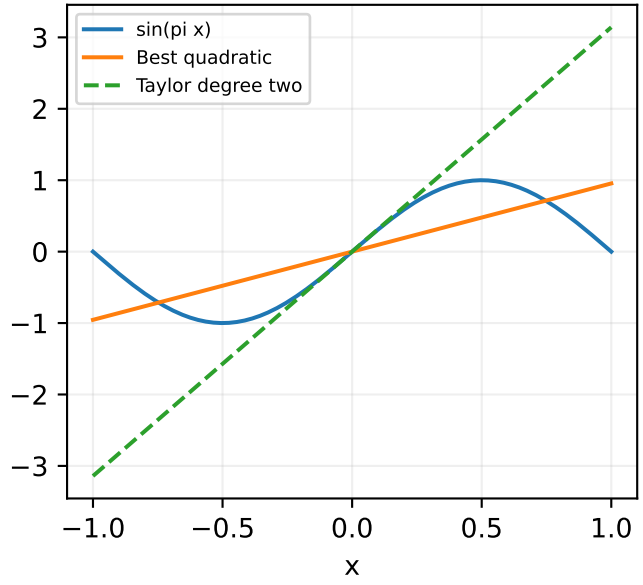

x squared
Unweighted projection
Weight 1 + x
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
x


x cubed
Legendre projection
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.0
−0.5
0.0
0.5
1.0
x


sin(pi x)
Best quadratic
Taylor degree two
3
2
1
0
−1
−2
−3
−1.0
−0.5
0.0
0.5
1.0
x